LESSON 9.3

Write Equations to Represent Relationships Between Two Variables

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation from a table or a graph to model the relationship between the dependent and independent variables.

WHY THIS MATTERS Miguel is interested in joining a recreation center. Recreation World charges the monthly fee shown on the flyer. How can Miguel determine the cost of being a member of Recreation World for any number of months?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
equation *Spanish: ecuación*	A mathematical sentence with an equals sign that shows two amounts are the same.	y = 5x
variable *Spanish: variable*	A letter that represents a quantity that can change, or vary.	x, y, c, h
rate *Spanish: tasa*	A fixed amount of one quantity for every one unit of the other quantity.	$3 for each pound
represent *Spanish: representar*	To show a relationship using a tool such as a table, a graph, or an equation.	The same membership shown as a table, a graph, AND c = 24.95m.
define the variables *Spanish: definir las variables*	To state exactly what quantity each letter in an equation stands for.	c = total cost
Constant *Spanish: Constante*	A number in an equation that never changes.	the 5 in y = 5x

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Write an equation for the pairs (1, 6), (2, 12), (5, 30).

1. 6 ÷ 1 = 6, 12 ÷ 2 = 6, 30 ÷ 5 = 6.
2. The rate is 6 every time.
3. Check: 6 × 5 = 30.

Answer: y = 6x

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A canoe rents for $15 per hour. Which quantity is independent — hours or total cost?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- In my equation, x stands for _____ and y stands for _____ .
- The constant rate is _____ because every y value is _____ .
- My equation is y = _____ , which means _____ .
- I checked my equation by substituting _____ for x.

Word bank equation · variable · rate · constant · define · substitute · times · per · table · row · relate · check

WATCH OUT!

A common mistake

Writing the equation with addition

LESSON 9.3

Write Equations to Represent Relationships Between Two Variables

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

In the equation c = 24.95m for the gym membership, what does m represent?

(A) The number of months of membership
(B) The total cost in dollars
(C) The monthly fee of $24.95
(D) The name of the gym

STEPS TO TRY
1. Which letters are variables here — and which quantity does each stand for?
2. c is the cost. What is left for m?
3. m counts the months.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Recreation World charges $24.95 per month. What is the total cost of 2 months of membership?

(A) $24.95
(B) $26.95
(C) $49.90
(D) $74.85

STEPS TO TRY

1. What does ONE month cost?
2. The second month adds another $24.95.
3. Multiply 24.95 by 2.

PER 1	

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

3 **Circle the letter of the best answer. Show how you know.**

A canoe rents for $15 per hour. Let h = hours and c = total cost. Which equation represents the relationship?

(A) c = 15h
(B) c = 15 + h
(C) h = 15c
(D) c = h − 15

STEPS TO TRY

1. Say it in words: the total cost equals $15 ___ the number of hours.
2. Every hour adds an equal $15 — which operation is that?
3. Test your choice with 2 hours: the total should be $30.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Using c = 15h, what is the total cost of a 3-hour canoe rental?

(A) $15 (B) $18

(C) $45 (D) $75

STEPS TO TRY

1. Replace h with 3 in the equation.
2. c = 15 × 3.
3. Skip-count: 15, 30, …

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

EXPLAIN YOUR THINKING

Both equations charge by a rate. What do m and h each stand for in their own situation, and why would c = 24.95m be wrong for the canoe?

SENTENCE STARTER In ___, ___ stands for ___, so the rate ___ means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help